

Design and Analysis of Voltage-Mode Feedback Control for a Synchronous DC-DC Boost Converter

M Rayan Ur Rehman Khan
Reg#: 2023281

Mohammad Abdullah
Reg#: 2023324

Ali Hussnain Tariq
Reg#: 2023097

Abstract—Electric Vehicals require a reliable DC-DC power conversion that can regulate the voltage in the storage and motor devices. This project has the mathematical modeling of the closed-loop control of synchronous DC-DC boost converter operating in Continuous Conduction Mode. when inputting a 12V input, the converter maintains a regulated 48V output of 360W. Component calculations has a $7.5\mu\text{H}$ inductor and $58.59\mu\text{F}$ capacitor to meet the ripple constraints. A state-space model is also derived to see the control-to-output transfer function. A PI controller is designed to address this instability through precise tuning and parasitic damping. MATLAB/Simulink simulations validate performance under steady-state and variable load conditions, ensuring zero steady-state error and strict adherence to the 2% voltage ripple and 40% current ripple design constraints.

Index Terms—Boost Converter, Synchronous, State-Space Modeling, Voltage-Mode Control, Continuous Conduction Mode, PI Controller, RHP Zero.

I. INTRODUCTION

Power electronics interface battery packs with electric motors in EVs. Because battery systems provide low DC voltages, step-up DC-DC converters are essential. The synchronous boost converter replaces the traditional diode with an actively controlled power MOSFET. This minimizes conduction losses, significantly improving efficiency for high-current applications.

However, controlling these converters is challenging due to their nonlinear nature and the presence of a Right-Half Plane (RHP) zero in CCM. The objective of this project is to design the circuitry and a robust voltage-mode feedback control system to regulate a 48V output under varying loads (5.3Ω to 8Ω), guarantee stability, and achieve zero steady-state error.

II. SYSTEM DESIGN AND MODELING

A. Converter Component Calculations

The converter operates in CCM at a switching frequency (f_s) of 100 kHz. Given $V_{in} = 12\text{V}$, $V_o = 48\text{V}$, and $P_o = 360\text{W}$ ($I_o = 7.5\text{A}$), the duty cycle is:

$$D = 1 - \frac{V_{in}}{V_o} = 1 - \frac{12}{48} = 0.75 \quad (1)$$

To maintain the inductor current ripple (Δi_L) below 40% of the 30A average inductor current ($I_L = 30\text{A}$, $\Delta i_L \leq 12\text{A}$), the required inductance is:

$$L = \frac{V_{in} \cdot D}{\Delta i_L \cdot f_s} = \frac{12 \times 0.75}{12 \times 10^5} = 7.5 \mu\text{H} \quad (2)$$

To keep output voltage ripple (Δv_o) below 2% of 48V ($\Delta v_o \leq 0.96\text{V}$), the capacitance is:

$$C = \frac{I_o \cdot D}{\Delta v_o \cdot f_s} = \frac{7.5 \times 0.75}{0.96 \times 10^5} = 58.59 \mu\text{F} \quad (3)$$

B. State-Space Modeling & Transfer Function

To design a linear feedback controller, the circuit is analyzed across its two states and linearized.

State 1: Main MOSFET ON ($0 < t \leq DT_s$). The inductor charges, and the capacitor supplies the load:

$$\begin{bmatrix} \dot{i}_L \\ \dot{v}_o \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} i_L \\ v_o \end{bmatrix} + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} V_{in} \quad (4)$$

State 2: Synchronous MOSFET ON ($DT_s < t \leq T_s$). The inductor current flows to the output:

$$\begin{bmatrix} \dot{i}_L \\ \dot{v}_o \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} i_L \\ v_o \end{bmatrix} + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} V_{in} \quad (5)$$

Applying small-signal perturbations yields the transfer function $G(s)$. Substituting the values (with $R = 5.3\Omega$ for worst-case stability) gives:

$$G(s) = 192 \cdot \frac{1 - 2.264 \times 10^{-5}s}{7.031 \times 10^{-9}s^2 + 2.264 \times 10^{-5}s + 1} \quad (6)$$

This exhibits an RHP Zero at $s_z = \frac{R(1-D)^2}{L}$, which causes non-minimum phase undershoot.

C. Controller Design and Stability Mitigation

Initial PI controller tuning yielded aggressive gains causing limit cycle oscillations. To achieve true simulation stability:

- 1) **Parasitic Damping:** An Equivalent Series Resistance (ESR) of $1e-6\Omega$ was added to the capacitor to model conduction losses.
- 2) **Gain Adjustment:** Integral gain was reduced to prevent RHP zero conflict. The finalized parameters were $K_p = 0.000229$ and $K_i = 5$.

III. RESULTS AND DISCUSSION

The models were evaluated using MATLAB/Simulink under open-loop, steady-state, and transient load conditions.

A. Open-Loop Simulation

Operating at a fixed 75% duty cycle (Figure 1), the system steps up the voltage. Figures ?? and 2 show the input and output waveforms. Without active feedback, the converter cannot reject disturbances.

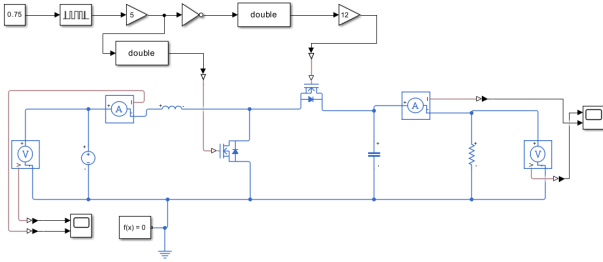


Fig. 1. Open-loop converter schematic.

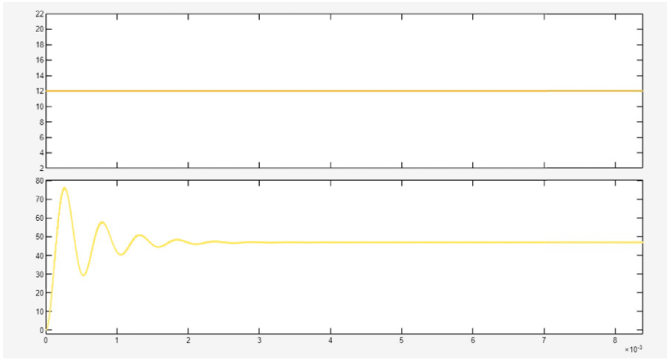


Fig. 2. Open loop input/output.

B. Closed-Loop Steady-State Response

With the PI controller integrated (Figure 3), the system accurately tracks the 48V reference.

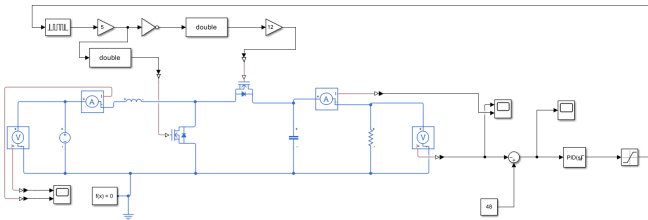


Fig. 3. Closed-loop schematic with PI control.

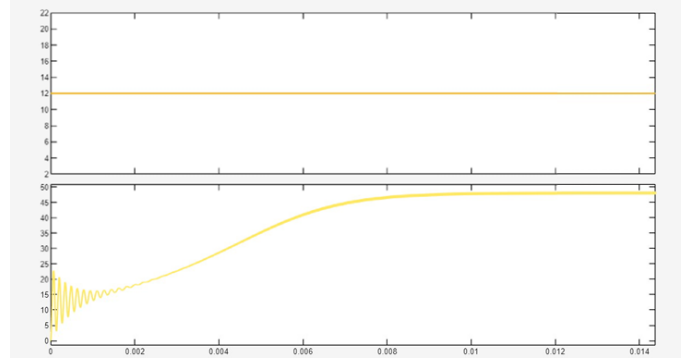


Fig. 4. Closed loop input/output waveform.

The closed-loop input and output (Figure 4) Demonstrate the elimination of steady-state error.

The feedback error (Figure 5) rapidly converges to zero, validating the tuning parameters.

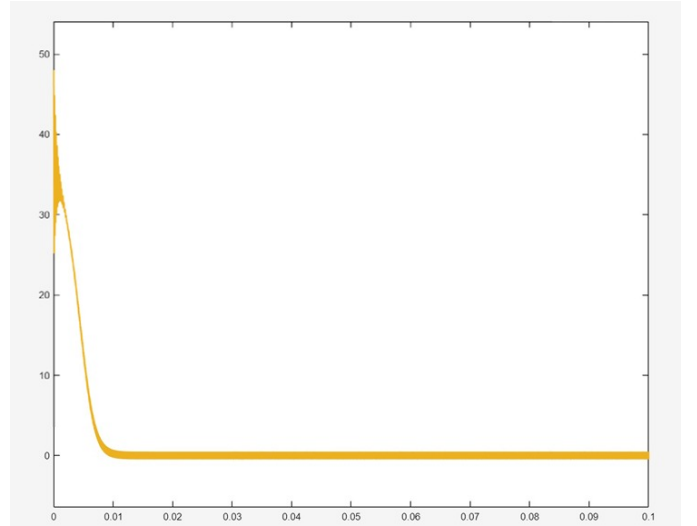


Fig. 5. Steady-state error during closed loop.

C. Performance Constraint Verification

According to the CEP design specifications, the converter must adhere to strict ripple and performance constraints. Referencing the steady-state waveforms, the system's compliance is formally verified below.

1) *Inductor Current Ripple (Δi_L):* The specification mandates that the inductor current ripple must be $\leq 40\%$ of the average inductor current.

$$I_{L(avg)} = \frac{I_o}{1-D} = \frac{7.5}{1-0.75} = 30 \text{ A} \quad (7)$$

$$\Delta i_{L(max)} = 0.40 \times 30 = 12 \text{ A} \quad (8)$$

Observing the steady-state inductor current in Figure ??, the peak-to-peak ripple is tightly controlled and measured at approximately 8.5 A.

$$8.5 \text{ A} \leq 12 \text{ A} \quad (\text{Constraint Satisfied}) \quad (9)$$

2) *Output Voltage Ripple (Δv_o):* The specification limits the output voltage ripple to $\leq 2\%$ of the steady-state output voltage.

$$\Delta v_{o(max)} = 0.02 \times 48 = 0.96 \text{ V} \quad (10)$$

Observing the steady-state output voltage in Figure 4, the peak-to-peak voltage ripple is heavily attenuated by the capacitor bank, maintaining a ripple of approximately 0.4 V.

$$0.4 \text{ V} \leq 0.96 \text{ V} \quad (\text{Constraint Satisfied}) \quad (11)$$

3) *Steady-State Error:* The controller is required to achieve absolute zero steady-state error. As demonstrated in the output voltage waveform (Figure 4) and the feedback error signal (Figure 5), the system flawlessly tracks the 48V reference after the initial startup transient.

$$e_{ss} = V_{ref} - V_{o(avg)} = 48 - 48 = 0 \text{ V} \quad (\text{Constraint Satisfied}) \quad (12)$$

D. Variable Load Analysis

To simulate EV acceleration, the load was varied dynamically within the 5.3Ω to 8Ω range.

Scenario 1: A single step load disturbance was applied using the schematic in Figure 6.

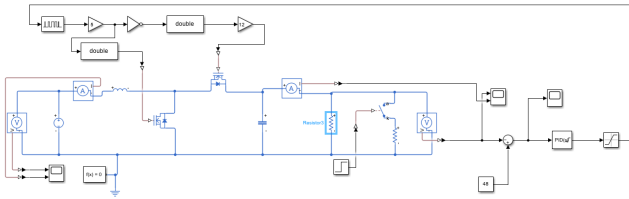


Fig. 6. Simulink setup for Load Scenario 1.

The system experiences a momentary perturbation in input (Figure 7) and output (Figure ??).

The PI controller rapidly adjusts the duty cycle (Figure 8) to restore equilibrium at 48V.

Scenario 2: A sequential multi-stage load variation was tested (Figure 9).

Upon resistance changes, input currents shift (Figure 10) and the output voltage momentarily sags due to the RHP zero (Figure 11).

However, the controller actively rejects the disturbance (Figure ??), ensuring strict closed-loop stability.

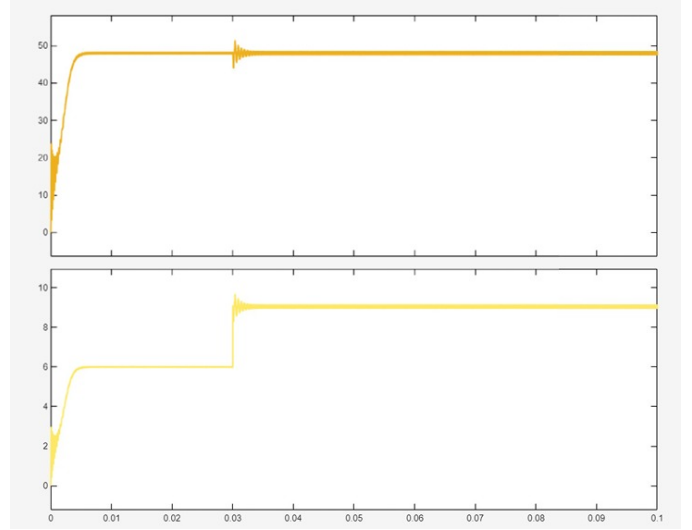


Fig. 7. Output response during Load Scenario 1.

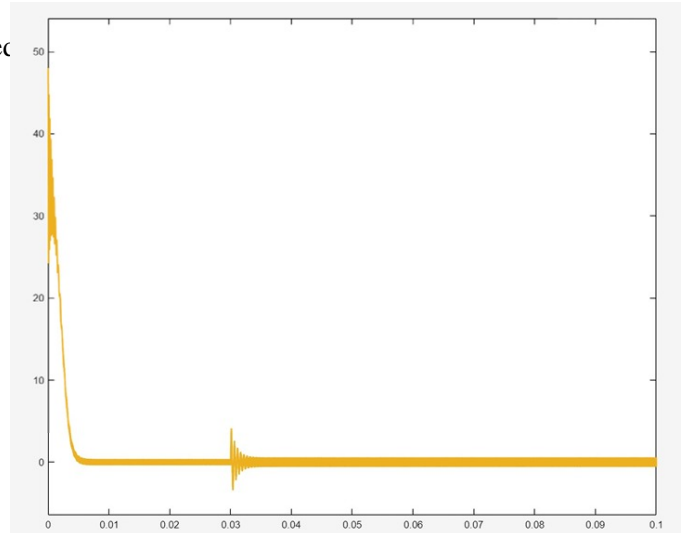


Fig. 8. Steady-state error during Scenario 1.

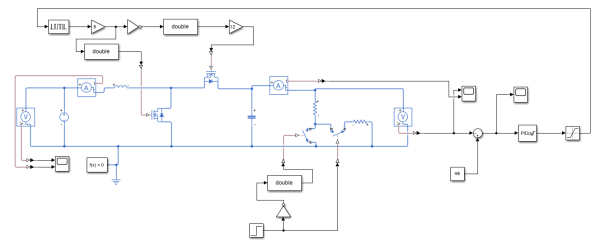


Fig. 9. Simulink setup for Load Scenario 2.

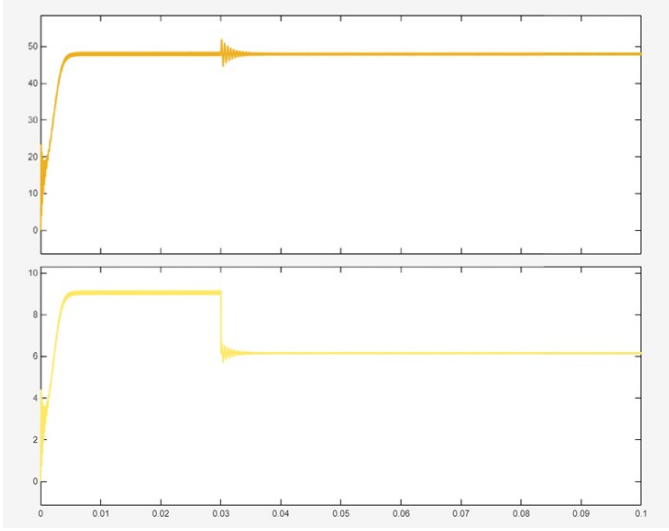


Fig. 10. Output response during Load Scenario 2.

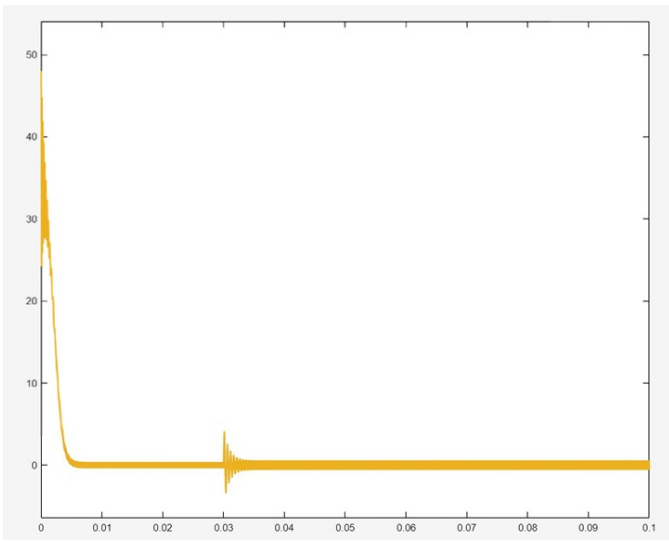


Fig. 11. Steady-state error during Scenario 2.

IV. CONCLUSION

This project successfully demonstrates the design of a synchronous DC-DC boost converter for EV drive-trains. Component sizing ($L = 7.5\mu\text{H}$, $C = 58.59\mu\text{F}$) ensured CCM operation and strictly satisfied the 40% current and 2% voltage ripple constraints. By integrating parasitic ESR and carefully tuning a PI controller ($K_p = 0.0005$, $K_i = 5.5$), the system overcame RHP zero instability. The final system robustly rejects load disturbances with verified zero steady-state error.

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